HIGH PERFORMANCE COMPUTING

Assignment - 6

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NUMA Effects and Bandwidth Measurement Objective: To observe how memory placement and NUMA effects influence performance and to understand bandwidth limitations.

Task 1: System NUMA Information lscpu | grep -i numa Observation: Number of NUMA nodes = _____

Code :

```
%%bash
# Install NUMA development libraries if not already installed
apt-get update -qq > /dev/null
apt-get install -y libnuma-dev > /dev/null

# Write the C code to a file
cat << EOF > numa_info.c
#include <stdio.h>
#include <numa.h>

int main() {

    if (numa_available() < 0) {
        printf("NUMA is not available on this system.\n");
        return 1;
    }

    int nodes = numa_max_node() + 1;
    int cpus = numa_num_configured_cpus();

    printf("NUMA is available on this system.\n");
    printf("Number of NUMA nodes = %d\n", nodes);
    printf("Number of CPUs = %d\n", cpus);

    for (int i = 0; i < nodes; i++) {
        printf("Node %d has %lld MB memory\n",
               i,
               numa_node_size(i, NULL) / (1024 * 1024));
    }

    return 0;
}
EOF
```

```
# Compile the C code
gcc numa_info.c -o numa_info -lnuma

# Execute the compiled program
./numa_info
```

OUTPUT:

```
NUMA is available on this system.
Number of NUMA nodes = 1
Number of CPUs = 2
Node 0 has 12975 MB memory

W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to
provide it (sources.list entry misspelt?)
```

Task 2: First Touch Memory Initialization File: numa_first_touch.c

#include <stdio.h>

#include <stdlib.h>

#include <omp.h>

#define N 10000000

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int main() { double *A = (double*)*malloc(sizeof(double)*N);

#pragma omp parallel for for (int i = 0; i < N; i++) A[i] = 1.0;

double sum = 0;

#pragma omp parallel for reduction(+:sum) for (int i = 0; i < N; i++) sum += A[i];

printf("Sum = %f\n", sum); free(A); return 0; } Compile & Run: gcc -O2 -fopenmp numa_first_touch.c -o numa_first_touch ./numa_first_touch

Code :

```
%%bash

cat << EOF > numa_first_touch.c
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>
#include <time.h>

#define N 10000000

int main() {
```

```c
    double *A = (double*)malloc(sizeof(double) * N);

    if (A == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    double start, end;

    // First Touch Initialization
    start = omp_get_wtime();

    #pragma omp parallel for
    for (int i = 0; i < N; i++) {
        A[i] = 1.0;
    }

    end = omp_get_wtime();
    printf("Initialization Time = %f seconds\n", end - start);

    // Summation
    double sum = 0.0;

    start = omp_get_wtime();

    #pragma omp parallel for reduction(+:sum)
    for (int i = 0; i < N; i++) {
        sum += A[i];
    }

    end = omp_get_wtime();

    printf("Sum = %f\n", sum);
    printf("Summation Time = %f seconds\n", end - start);

    free(A);
    return 0;
}
```

OUTPUT :

```
EOF

gcc -O2 -fopenmp numa_first_touch.c -o numa_first_touch -lm
./numa_first_touch

Initialization Time = 0.029790 seconds
Sum = 10000000.000000
Summation Time = 0.008760 seconds
```

./numa_first_touch

Task 3: Conceptual Questions

1. What is first-touch policy?
1. Does NUMA affect correctness or performance?
2. Why is memory bandwidth important in NUMA systems?

%%bash

Code :

```
cat << EOF > memory_bandwidth.c
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define N 50000000

int main() {

    double *A = (double*)malloc(sizeof(double) * N);
    double *B = (double*)malloc(sizeof(double) * N);

    if (A == NULL || B == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    // Initialize arrays
    #pragma omp parallel for
    for (int i = 0; i < N; i++) {
        A[i] = 1.0;
        B[i] = 2.0;
    }

    double start = omp_get_wtime();

    #pragma omp parallel for
    for (int i = 0; i < N; i++) {
        A[i] = A[i] + B[i];
    }

    double end = omp_get_wtime();

    double time = end - start;

    double bytes_accessed = 3 * sizeof(double) * N;
    double bandwidth = (bytes_accessed / time) / 1e9;
```

```
    printf("Execution Time = %f seconds\n", time);
    printf("Estimated Memory Bandwidth = %f GB/s\n", bandwidth);

    free(A);
    free(B);

    return 0;
}
```

OUTPUT :

```
EOF

gcc -O2 -fopenmp memory_bandwidth.c -o memory_bandwidth -lm
./memory_bandwidth

Execution Time = 0.091972 seconds
Estimated Memory Bandwidth = 13.047410 GB/s
```